

Macroeconomics II, Lecture X:

Diamond-Mortensen-Pissarides: Efficiency and Dynamics

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- DMP: a GE theory of unemployment
 - ▶ emphasizes vacancy creation and job destruction as key mechanisms
- Defining elements: matching function + wage-setting rule + free-entry condition
- Last lecture: steady-state equilibrium and comparative statics
- Today: welfare and dynamics
- We will focus on the vanilla DMP model with exogenous separations, but unless noted otherwise, all results extend to the endogenous-separation model

Recap: equilibrium characterization with exogenous separations

- Summary: the equilibrium steady state $\{w, \theta, u, v\}$ is characterized by

$$\text{JC:} \quad w = y - \frac{c(r + \sigma)}{\lambda_v(\theta)}$$

$$\text{WC:} \quad w = (1 - \gamma)b + \gamma(y + c\theta)$$

$$\text{BC:} \quad u = \frac{\sigma}{\sigma + \lambda_u(\theta)}$$

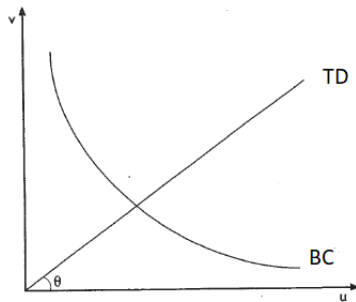
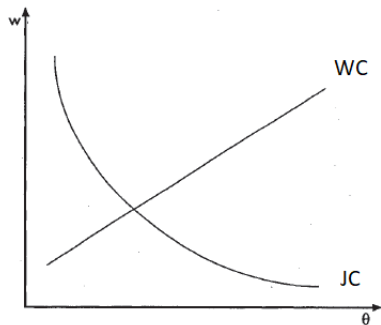
$$\text{TD:} \quad v = \theta u$$

- Note: the system is *block recursive*: we can solve for θ by combining the first two equations

$$\frac{c}{\lambda_v(\theta)} = \frac{(1 - \gamma)(y - b)}{r + \sigma + \lambda_u(\theta)\gamma}$$

- Given θ , we can solve for u and v from the last two equations

Recap: graphical view of equilibrium



- ① Efficiency
- ② Dynamics
 - ▶ Steady-state approximation of dynamic equilibrium
 - ▶ Unemployment volatility puzzle
- ③ Summary and discussion

Efficiency

- Is the DMP equilibrium efficient?
 - ▶ Clearly not, in the usual sense
 - ▶ An unconstrained social planner would ignore the search frictions and allocate all workers to firms at all points in time
- Is the DMP equilibrium **constrained efficient**?
 - ▶ Constrained efficiency: would a social planner who can choose the actions of all agents, but still faces the same frictions, choose differently from agents in the decentralized equilibrium?
- Remember: the only choice in DMP is whether to post a vacancy
- Reformulated question: would the creation of more vacancies increase or decrease welfare?
- In general, two externalities arise when a firm creates an additional vacancy
 - ▶ Positive *thick market externality*: workers will find jobs faster
 - ▶ Negative *congestion externality*: other firms will find workers slower
 - ▶ Note: these externalities arise due to the matching function

Social planner problem

- To find out, we need to compare the social planner's solution to the decentralized solution
- Social planner problem (with equal Pareto weights):

$$\begin{aligned} \max_v \quad & \int_0^{\infty} e^{-rt} ((1-u)y + ub - vc) dt \\ \text{s.t.} \quad & \dot{u} = \sigma(1-u) - \lambda_u \left(\frac{v}{u}\right) u \end{aligned}$$

with u as the state variable and v as the control

- We can reformulate as

$$\begin{aligned} \max_{\theta} \quad & \int_0^{\infty} e^{-rt} ((1-u)y + ub - \theta uc) dt \\ \text{s.t.} \quad & \dot{u} = \sigma(1-u) - \lambda_u(\theta) u \end{aligned}$$

with u as the state variable and θ as the control

Social planner problem: solution

- Introduce the co-state μ and form the Hamiltonian:

$$H(u, \theta, \mu, t) = e^{-rt} [(y(1-u) + bu - \theta uc) + \mu(\sigma(1-u) - \lambda_u(\theta)u)]$$

- From Pontryagin's principle, we have the optimality conditions:

$$\begin{aligned}\frac{dH}{du} &= -\dot{\mu} \\ \frac{dH}{d\theta} &= 0\end{aligned}$$

and thus

$$\begin{aligned}-y + b - \theta c - \mu(\sigma + \lambda_u(\theta)) &= -\dot{\mu} \\ -c - \mu\lambda'_u(\theta) &= 0\end{aligned}$$

- μ = the value of having one more unemployed worker. $-\mu$ = the expected cost of matching that unemployed worker with a new vacancy:

$$-\mu = \frac{c}{\lambda'_u(\theta)}$$

- Evaluate in steady state $\dot{\theta} = 0$ and work out the algebra to find

$$\frac{c}{\lambda_v(\theta)} = \frac{(1 - \epsilon_{m,u}(\theta))(y - b)}{r + \sigma + \lambda_u(\theta)\epsilon_{m,u}(\theta)}$$

where

$$\epsilon_{m,u} = \frac{dM(u, v)}{du} \frac{u}{M(u, v)} = 1 - \frac{\theta \lambda'_u(\theta)}{\lambda_u(\theta)}$$

- Compare to decentralized solution

$$\frac{c}{\lambda_v(\theta)} = \frac{(1 - \gamma)(y - b)}{r + \sigma + \lambda_u(\theta)\gamma}$$

Is the decentralized equilibrium efficient?

- Hosios (REStud 1990): the decentralized equilibrium is efficient iff $\gamma = \epsilon_{m,u}(\theta)$!
- If $\epsilon_{m,u} < \gamma$, workers receive too large a share of the surplus, leading to too little vacancy creation (the thick market externality dominates and unemployment is too high)
- If $\epsilon_{m,u} > \gamma$, firms receive too large a share of the surplus and spend too many costly resources on creating vacancies (the congestion externality dominates and unemployment is too low)
- If $\epsilon_{m,u} = \gamma$, the social value of creating an additional vacancy equals the private gain to the firm

- In this model, there is no reason why we should expect $\epsilon_{m,u} = \gamma$
- Random matching implies generically inefficient equilibria, leaving room for welfare-improving government interventions
- Primary motivation for models with [directed search](#): they introduce competitive forces that ensure an efficient benchmark (Moen, JPE 1997; Menzio and Shi, JPE 2011)
- See also Névo (2025)

Dynamics

The question

- So far, our analysis has only concerned the steady state
- **Can DMP explain the cyclical unemployment?**
- This question refers to the dynamic response of the unemployment rate to an underlying shock
- A natural candidate: shocks to productivity y

- Strategy: calibrate model primitives to match long-run moments, feed in shocks that fit the detrended processes of TFP and separation rates, and evaluate unemployment dynamics
 - ▶ Just as you would evaluate an RBC or New Keynesian model of business-cycle dynamics in output

- A “proper” evaluation requires us to

- ① specify a process for the aggregate productivity shock, e.g., shocks arrive at rate z
- ② describe how the value functions change with the aggregate state, e.g.,

$$rU_y = b + \lambda_u(\theta_y)(W_y - U_y) + zE_{y'|y}(U_{y'} - U_y)$$

- ③ solve the model numerically

- However, insights about the dynamics of DMP can be gained by simple comparative statics
- This is because the dynamics in DMP (and in the data) are fast: with realistic job-finding rates, the model converges almost instantaneously to steady state

Fast dynamics: a steady-state approximation

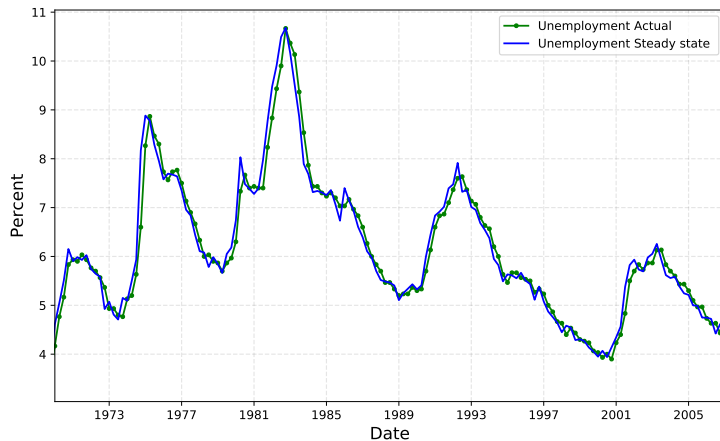
- Consider the formula for steady-state unemployment:

$$u_{ss} = \frac{\sigma}{\sigma + \lambda_{u,t}}$$

- At any given t , there exists a steady-state unemployment rate $u_{ss,t}$ associated with the current flow rates

$$u_{ss,t} = \frac{\sigma_t}{\sigma_t + \lambda_{u,t}}$$

- How does $u_{ss,t}$ compare with actual unemployment u_t ?



Own calculations based on quarterly data. Unemployment data: FRED. Transition rate data: Shimer's website

Fast dynamics II

- What's going on?
- Consider the law of motion for unemployment

$$\dot{u} = \sigma(1 - u_t) - \lambda_u(\theta)u_t$$

- This is a **first-order linear ordinary differential equation**
- General solution:

$$u_t = Ce^{-(\sigma + \lambda_u(\theta))t} + \frac{\sigma}{\sigma + \lambda_u(\theta)}$$

for some constant C

- For any positive σ, λ_u : $u_t \rightarrow u_{ss} = \frac{\sigma}{\sigma + \lambda_u(\theta)}$
- Larger $\sigma, \lambda_u \rightarrow$ faster convergence
- What is the half-life of **out-of-steady-state unemployment** ? That is, which T solves the following equation?

$$\frac{u_T - u_{ss}}{u_0 - u_{ss}} = \frac{1}{2}$$

Fast dynamics III

- Steady state unemployment:

$$u_{ss} = \frac{\sigma}{\sigma + \lambda_u(\theta)}$$

- Half-life given by

$$\begin{aligned}\frac{u_T - u_{ss}}{u_0 - u_{ss}} &= \frac{1}{2} \\ \frac{Ce^{-(\sigma + \lambda_u(\theta))T} + \frac{\sigma}{\sigma + \lambda_u(\theta)} - \frac{\sigma}{\sigma + \lambda_u(\theta)}}{Ce^{-(\sigma + \lambda_u(\theta)) \cdot 0} + \frac{\sigma}{\sigma + \lambda_u(\theta)} - \frac{\sigma}{\sigma + \lambda_u(\theta)}} &= \frac{1}{2} \\ e^{-(\sigma + \lambda_u(\theta))T} &= \frac{1}{2}\end{aligned}$$

- That is, $T = -\frac{1}{\sigma + \lambda_u} \log \frac{1}{2}$
- US monthly transition rates $\sigma \approx 0.03$, $\lambda_u \approx 0.6 \Rightarrow T = 1.1$ months
- Since convergence is so fast: $u_{ss}(\sigma_t, \lambda_{ut}) \approx u_t$

Solving dynamic DMP without simulations?

- Lesson: given sufficiently high transition rates, the labor market converges to steady state very fast
- Implication: Given that DMP is calibrated to match these high transition rates, the DMP equilibrium converges to steady state very fast
- Consider a dynamic DMP model in which a simulated sequence of productivity $\{y_0, y_1, y_2, \dots\}$ maps into a sequence of unemployment $\{u_0, u_1, u_2, \dots\}$
- Since convergence to steady state is almost immediate, we can approximate the unemployment sequence with $\{u_{ss}(y_0), u_{ss}(y_1), u_{ss}(y_2), \dots\}$
- That is, we can simply do comparative statics of steady-state unemployment u with respect to y !

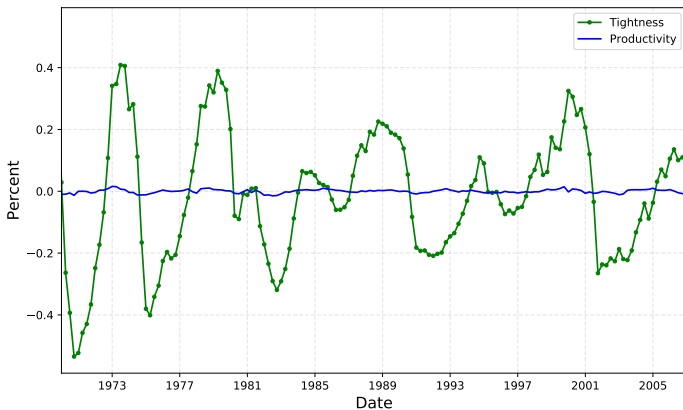
Recap: steady state equilibrium

- Recall: the steady-state equilibrium is given by

$$\frac{c}{\lambda_v(\theta)} = \frac{(1 - \gamma)(y - b)}{r + \sigma + \lambda_u(\theta)\gamma}$$
$$u = \frac{\sigma}{\sigma + \lambda_u(\theta)}$$

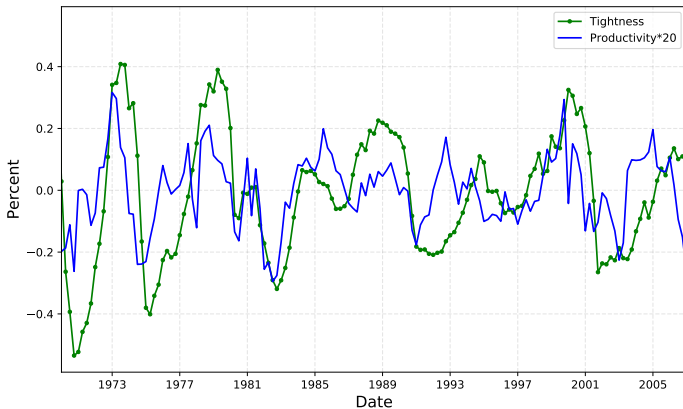
- Given that the matching function is (approximately) correct, the model response of u to a y shock matches the data if the elasticity of θ with respect to y matches the data

Tightness-productivity elasticity in the data



Own calculations using HP-filtered quarterly data. OECD labor productivity measure downloaded from FRED. Tightness constructed using BLS unemployment from FRED and vacancy data (Help-Wanted Index) from Regis Barnichon's website.

Tightness-productivity elasticity in the data, rescaled



Own calculations using HP-filtered quarterly data. OECD labor productivity measure downloaded from FRED. Tightness constructed using BLS unemployment from FRED and vacancy data (Help-Wanted Index) from Regis Barnichon's website.

Can DMP explain unemployment volatility? Shimer (AER 2005)

- Empirical elasticity $\epsilon_{\theta,y} \approx 20$
- DMP elasticity $\epsilon_{\theta,y}$?
- Job creation curve in $\{y, \theta\}$ -space (using $\lambda_u(\theta) = \theta\lambda_v(\theta)$):

$$\frac{c}{\lambda_v(\theta)} = \frac{(1 - \gamma)(y - b)}{r + \sigma + \lambda_u(\theta)\gamma}$$

- Total differentiation to find (Do on whiteboard):

$$\epsilon_{\theta,y} \equiv \frac{y}{\theta} \frac{\partial \theta}{\partial y} = \frac{y}{y - b} \frac{r + \sigma + \gamma\lambda_u(\theta)}{(r + \sigma)(1 - \epsilon_{\lambda_u,\theta}) + \gamma\lambda_u(\theta)}$$

where $\epsilon_{\lambda_u,\theta} \equiv \frac{\theta}{\lambda_u(\theta)} \lambda_u'(\theta)$ is the elasticity of $\lambda_u(\theta)$ with respect to θ

Can DMP explain unemployment volatility? Shimer (AER 2005)

$$\epsilon_{\theta,y} = \frac{y}{y-b} \frac{r + \sigma + \gamma \lambda_u(\theta)}{(r + \sigma)(1 - \epsilon_{\lambda_u, \theta}) + \gamma \lambda_u(\theta)}$$

- Shimer's (standard) calibration, quarterly frequency:
 - ▶ Normalize $y = 1$
 - ▶ Take outside estimates $\sigma = 0.1, r = 0.012, b = 0.4$
 - ▶ Estimate a Cobb-Douglas matching function with $A = 1.34, \alpha = 0.72, \epsilon_{\lambda_u, \theta} = 1 - \alpha$
 - ▶ Set $c = 0.2$ to match steady state tightness θ

- If $\gamma = \alpha$ (Hosios condition): $\lambda_u(\theta) = 2.13$ and

$$\epsilon_{\theta,y} = \frac{y}{y-b} \times \frac{r + \sigma + \alpha \lambda_u(\theta)}{(r + \sigma)(1 - \epsilon_{\lambda_u, \theta}) + \alpha \lambda_u(\theta)} = 1.7$$

- If $\gamma = 0$:

$$\epsilon_{\theta,y} = \frac{y}{y-b} \times \frac{1}{1 - \epsilon_{\lambda_u, \theta}} = 3.5$$

- The model comes nowhere near the data

- Why does DMP fail to generate unemployment volatility?
- Shimer suggests intuition:
 - ▶ $y \uparrow \Rightarrow \text{Value of vacancy} \uparrow$
 - ▶ Free entry and $v \uparrow \Rightarrow \text{Tightness} \uparrow$
 - ▶ $\lambda_u(\theta) \uparrow \Rightarrow \text{Value of unemployment} \uparrow$
 - ▶ Worker's outside option $\uparrow \Rightarrow \text{Wages} \uparrow$
 - ▶ Value of vacancy $\downarrow \Rightarrow \text{Equilibrium response small}$
- Summarizing: the equilibrium wage response through Nash bargaining inhibits firms' incentive to create vacancies

The Shimer puzzle

- Result: a reasonable calibration of the basic DMP model cannot explain observed unemployment volatility
- What about endogenous separations? Same problem, and actually creates another one:
 - ▶ Consider a positive separation shock (as implied by a negative productivity shock)
 - ▶ Higher u , but also lower θ
 - ▶ $\Rightarrow$ induces more vacancy creation
 - ▶ $\Rightarrow$ positive correlation between vacancies and unemployment!
- However, this result relies heavily on the free-entry condition, as we will see below

The Shimer puzzle: resolutions

- Hall (AER 2005): Ad hoc wage rigidity
- Hagedorn-Manovskii (AER 2008): alternative calibration
- Hall (AER 2017): discount factor shocks
- Coles-Kelishomi (AEJ Macro 2018): sluggish vacancy creation

The Shimer puzzle: resolutions

- Hall (AER 2005): Ad hoc wage rigidity
- **Hagedorn-Manovskii (AER 2008): alternative calibration**
- Hall (AER 2017): discount factor shocks
- **Coles-Kelishomi (AEJ Macro 2018): separation shocks and sluggish vacancy creation**

- Hagedorn and Manovskii argue: the previous literature calibrated the DMP model incorrectly; if you do it right, there is no puzzle
- Again, tightness-productivity elasticity with Nash bargaining:

$$\epsilon_{\theta,y} = \frac{y}{y-b} \frac{r + \sigma + \gamma \lambda_u(\theta)}{(r + \sigma)(1 - \epsilon_{\lambda_u, \theta}) + \gamma \lambda_u(\theta)}$$

- If b is high relative to y , the model can generate substantial unemployment fluctuations
- Why is b the key parameter? Let's investigate

Hagedorn and Manovskii's argument

- Intuition: To generate a strong vacancy creation response to a productivity shock, two criteria need to be satisfied
 - ① The productivity increase must not be fully absorbed by an increase in wages
 - ② The initial level of profits needs to be small
- Why does the initial level of profits matter?
 - ▶ The elasticity of vacancy creation with respect to productivity depends on the elasticity of profits with respect to productivity
 - ▶ Suppose wages are fixed and study a 1 percent productivity increase;
 $y_0 = 1 \rightarrow y_1 = 1.01$
 - ▶ Suppose $w = 0.99$:

$$\epsilon_{\pi,y} = 100 * \frac{(1.01 - 0.99) - (1 - 0.99)}{(1 - 0.99)} = 100\%$$

- ▶ Suppose $w = 0.01$:

$$\epsilon_{\pi,y} = 100 * \frac{(1.01 - 0.01) - (1 - 0.01)}{1 - 0.01} = 1\%$$

Hagedorn and Manovskii's argument II

- When is initial level of profits (wages) small (large) with Nash Bargaining?
- Look at wage curve:

$$\begin{aligned}w &= (1 - \gamma)b + \gamma(y + c\theta) \\ &= b + \gamma(y - b) + \gamma c\theta\end{aligned}$$

Either γ or b must be high

- Why then was it only b that seemed to have a large impact on the tightness-productivity elasticity?

$$\epsilon_{\theta,y} = \frac{y}{y - b} \frac{r + \sigma + \gamma\lambda_u(\theta)}{(r + \sigma)(1 - \epsilon_{\lambda_u,\theta}) + \gamma\lambda_u(\theta)}$$

- High γ also implies volatile wages with respect to $y \rightarrow$ the level and volatility effects on vacancy elasticity approximately cancel each other out

Hagedorn and Manovskii's argument III

- Main question: do the data support a high b ?
- Shimer did not set his parameters to be consistent with the cyclicalities of wages. Instead, he set $b = 0.4$, $\gamma = \alpha$, and adjusted the vacancy posting cost to match steady-state tightness
- H&M stress:
 - ▶ In the data, wages are moderately procyclical, $\epsilon_{w,y} = 0.45$
 - ▶ taking both labor and capital cost of vacancy posting into account, $c \approx 0.6$
- For a given b and vacancy posting cost c , $\epsilon_{w,y}$ pins down γ
- H&M match this moment and find a much lower $\gamma = 0.05$
- With lower γ , b must be higher to match the same steady state tightness θ
- H&M find $b = 0.95$

Hagedorn and Manovskii's calibration

- With these numbers, the tightness-productivity elasticity becomes:

$$\epsilon_{\theta,y} = \frac{y}{y-b} \frac{r + \sigma + \gamma \lambda_u(\theta)}{(r + \sigma)(1 - \epsilon_{\lambda_u, \theta}) + \gamma \lambda_u(\theta)} = 20.6$$

- Does this make economic sense? H&M: yes!
- b is really the total utility flow of being unemployed: unemployment benefits, leisure value, home production value, etc.
- A larger model of household labor decisions will equate the marginal value of search for a job with the marginal cost of giving up time for nonmarket activities.
- If the value of unemployment is not close to that of being employed, it is difficult to explain why unemployed people do not search harder for jobs
- This is a controversial argument: if the welfare loss of being unemployed is so small, why do we seemingly care so much about it?

Comment: wage levels and wage rigidity

- Shimer's intuition was that the failure to match unemployment volatility reflected the flexibility of wages in the model
- H&M showed that this intuition is partly wrong, or at least, incomplete
 - ▶ Yes, the productivity increase must not be fully absorbed by an increase in wages in response to the shock
 - ▶ but, given that it is not fully absorbed, the important thing is whether the steady-state level of profits/wages is high or low relative to productivity
 - ▶ Elaborated in Ljungqvist-Sargent (AER 2017)
- Nevertheless, much of the literature has focused on wage rigidity
 - ▶ Hall (AER 2005): DMP with an ad hoc fully rigid wage shows that the model matches unemployment cyclicalities if the steady-state wage is high enough
 - ▶ Pissarides (Ecmtra 2009): it is only the wage of new hires that is allocative in the model; wage rigidity among incumbents is unimportant
 - ▶ Mixed evidence on the extent of wage rigidity for new hires in the literature, see e.g., Kudlyak (JME 2014), Grigsby-Hurst-Yildirmaz (AER 2021), Hazell-Taska (AER 2024)
 - ▶ Carlsson-Westermarck (AEJ Macro 2022): with endogenous separations, rigidity among incumbents also matters

- CK's argument: unemployment variability can be explained with separation shocks if we relax free entry
- Free entry is reasonable in the long run, but in the short run there are all kinds of barriers
- Extend the vanilla DMP model with stochastic entry costs κ

$$\kappa \sim H(\cdot)$$

- $\Rightarrow$ all firms drawing $\kappa < V_t$ create a vacancy. Vacancy creation rate: $\iota_t = H(V_t)$
- With $H(x) = x^\xi$, vacancy creation ι_t has a constant elasticity with respect to vacancy values:

$$\iota_t = V_t^\xi$$

- Note: Free entry means $\xi \rightarrow \infty$

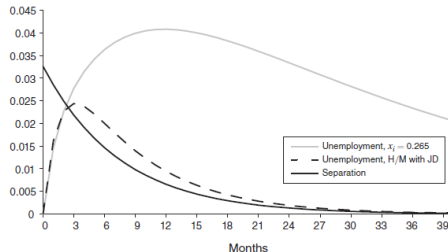


FIGURE 2. IMPULSE RESPONSE OF UNEMPLOYMENT TO A SEPARATION SHOCK

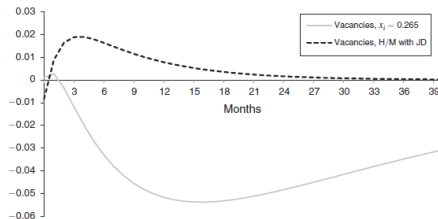


FIGURE 3. IMPULSE RESPONSE OF VACANCIES TO A SEPARATION SHOCK

- With sluggish vacancy creation, newly separated deplete the current vacancy stock
 - ▶ Negative correlation between vacancy stock and unemployment, as in the data
 - ▶ $\Rightarrow$ Increase in separations depresses the job-finding rate, double effect on unemployment
 - ▶ Same logic with endogenous separations and productivity shocks
- Growing literature on implications, microfoundations, and empirics:
 - ▶ Schoefer-Mercan-Sedlacek (AEJ Macro 2023): sluggish entry due to imperfect substitutability
 - ▶ Engbom (2021): sluggish entry due to compositional effects over the cycle
 - ▶ Broer-Druedahl-Harmenberg-Öberg (IER 2025): implications for unemployment-risk dynamics

- DMP: a GE theory of unemployment levels and dynamics
 - ▶ emphasizes vacancy creation as the key economic mechanism
- Defining elements: matching function + wage-setting rule + free-entry condition
- Generates reasonable predictions, explored by growing empirical literature
- Key question: can DMP explain unemployment volatility?
- Maybe, but still not fully understood

Search models: what we have not covered

- The theory of money
 - ▶ Goods markets are characterized by the absence of *double coincidence of wants*
 - ▶ Search process without mediating transaction technology is very costly
 - ▶ Useless assets, such as fiat money, can have positive value because they mediate trade
 - ▶ See Williamson and Wright (Handbook ME 2010)
- Liquidity in financial markets
 - ▶ Some financial markets are close to the Walrasian ideal. Others, such as the market for mortgage-backed securities, operate with *over-the-counter* trading
 - ▶ Search theory characterizes trading volume, bid-ask spreads, and how the supply of buyers might collapse during crises
 - ▶ See Lagos and Rocheteau (Ecmtra 2009)
- Family economics
 - ▶ Marriage is a matching process under costly search
 - ▶ See Shimer and Smith (Ecmtra 2000)

Search models: what we have not covered: Epidemiology

- The baseline model of disease spread is a random matching model - the SIR model
- Given a number of Susceptibles S and a number of infected I , assume that the number of newly infected is given by matching function $M = \beta SI$
- Laws of motion

$$\begin{aligned}\dot{S} &= -\beta SI \\ \dot{I} &= \beta SI - \gamma I \\ \dot{R} &= \gamma I\end{aligned}$$

where R is the number of dead/recovered

- SIR offers a way to forecast disease spread and study lockdown policies
- In the epidemiology literature, there is little treatment of optimizing behavior and incentives
 - ▶ In other words, β is taken as a parameter and not an endogenous variable
- Eichenbaum-Rebelo-Trabant (RFS 2021): $\beta = \beta(C_t, H_t)$, integrate SIR with standard RBC economy
 - ▶ Many papers are currently extending this analysis in interesting ways